Linux User Management & Security Automation System (TECHNOVA Case Study)

Company: TECHNOVA

Linux Administration and Security Automation Case Study

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Focus Area: User Management | RBAC | Password Policy | Backup Automation | Account Lifecycle

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1) Problem Statement

1.1 Organization Scenario

TECHNOVA is a multi-department organization where multiple users need secure and structured access to Linux system resources. Because every department has different users, directories, and access needs, manual administration can become risky, repetitive, and difficult to audit.

1.2 Department-wise Infrastructure

Department	Total PCs
HR	10
Finance	15
Server	5
Admin	8
Sales	23
Interns	20
IT	12

1.3 User Naming Convention

Department	Username Range
HR	hr01 to hr10
Finance	fin01 to fin15
Server	srv01 to srv05
Admin	adm01 to adm08
Sales	sales01 to sales23
Interns	int01 to int20
IT	it01 to it12

1.4 Administrative Challenges

- User management across multiple departments
- Security policy enforcement for all users
- Backup handling and recovery planning
- Role-based access control for department directories
- Automation of repeated Linux administration tasks

1.5 Required Work

12. Create users
13. Assign passwords
14. Create groups
15. Manage user directories
16. Apply password policy
17. Backup user data
18. Set permissions according to role
19. Automation using cron jobs
20. Account expiry for interns
21. Automatic password expiry reminder
22. Inactive account locking

23. Force password change on first login
24. Service management
25. Create welcome/login banner
26. Package management
27. Default file setup using /etc/skel

2) Objective

To create department-wise Linux groups, add users, set passwords, verify users and groups, apply password policy, assign directory permissions, and configure backup automation for TECHNOVA.

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3) Department-wise User and Group Creation

This section contains the actual Linux commands used to create groups, add users, set passwords, and verify the implementation.

3.1 Department 1: Admin

Create Group

```
groupadd admin
```

Create Users

```
useradd -G admin adm01
useradd -G admin adm02
useradd -G admin adm03
useradd -G admin adm04
useradd -G admin adm05
useradd -G admin adm06
useradd -G admin adm07
useradd -G admin adm08
```

Set Passwords

```
passwd adm01
# enter: adm01

passwd adm02
# enter: adm02

passwd adm03
# enter: adm03

passwd adm04
# enter: adm04

passwd adm05
# enter: adm05
```

```
passwd adm06
# enter: adm06

passwd adm07
# enter: adm07

passwd adm08
# enter: adm08
```

Verify

```
ls /home
cat /etc/passwd | grep adm
cat /etc/group | grep admin
```

3.2 Department 2: Server

Create Group

```
groupadd server
```

Create Users

```
useradd -G server srv01
useradd -G server srv02
useradd -G server srv03
useradd -G server srv04
useradd -G server srv05
```

Set Passwords

```
passwd srv01
# enter: srv01

passwd srv02
# enter: srv02

passwd srv03
# enter: srv03

passwd srv04
# enter: srv04

passwd srv05
# enter: srv05
```

Verify

```
ls /home
cat /etc/passwd | grep srv
cat /etc/group | grep server
```

3.3 Department 3: Sales

Create Group

```
groupadd sales
```

Create Users

```
useradd -G sales sales01
useradd -G sales sales02
useradd -G sales sales03
useradd -G sales sales04
useradd -G sales sales05
useradd -G sales sales06
useradd -G sales sales07
useradd -G sales sales08
useradd -G sales sales09
useradd -G sales sales10
useradd -G sales sales11
useradd -G sales sales12
useradd -G sales sales13
useradd -G sales sales14
useradd -G sales sales15
useradd -G sales sales16
useradd -G sales sales17
useradd -G sales sales18
useradd -G sales sales19
useradd -G sales sales20
useradd -G sales sales21
useradd -G sales sales22
useradd -G sales sales23
```

Set Passwords

```
passwd sales01
# enter: sales1

passwd sales02
# enter: sales2

passwd sales03
# enter: sales3

passwd sales04
# enter: sales4

passwd sales05
# enter: sales5

passwd sales06
# enter: sales6

passwd sales07
# enter: sales7

passwd sales08
```

```
# enter: sales8

passwd sales09
# enter: sales9

passwd sales10
# enter: sales10

passwd sales11
# enter: sales11

passwd sales12
# enter: sales12

passwd sales13
# enter: sales13

passwd sales14
# enter: sales14

passwd sales15
# enter: sales15

passwd sales16
# enter: sales16

passwd sales17
# enter: sales17

passwd sales18
# enter: sales18

passwd sales19
# enter: sales19

passwd sales20
# enter: sales20

passwd sales21
# enter: sales21

passwd sales22
# enter: sales22

passwd sales23
# enter: sales23
```

Verify

```
ls /home
cat /etc/passwd | grep sales
cat /etc/group | grep sales
```

3.4 Department 4: HR

Create Group

```
groupadd hr
```

Create Users

```
useradd -G hr hr1
useradd -G hr hr2
useradd -G hr hr3
useradd -G hr hr4
useradd -G hr hr5
useradd -G hr hr6
useradd -G hr hr7
useradd -G hr hr8
useradd -G hr hr9
useradd -G hr hr10
```

Set Passwords

```
passwd hr1
# enter: hr01

passwd hr2
# enter: hr02

passwd hr3
# enter: hr03

passwd hr4
# enter: hr04

passwd hr5
# enter: hr05

passwd hr6
# enter: hr06

passwd hr7
# enter: hr07

passwd hr8
# enter: hr08

passwd hr9
# enter: hr09

passwd hr10
# enter: hr010
```

Verify

```
ls /home
cat /etc/passwd | grep hr
cat /etc/group | grep hr
```

3.5 Department 5: Finance

Create Group

```
groupadd finance
```

Create Users

```
useradd -G finance fin01
useradd -G finance fin02
useradd -G finance fin03
useradd -G finance fin04
useradd -G finance fin05
useradd -G finance fin06
useradd -G finance fin07
useradd -G finance fin08
useradd -G finance fin09
useradd -G finance fin10
useradd -G finance fin11
useradd -G finance fin12
useradd -G finance fin13
useradd -G finance fin14
useradd -G finance fin15
```

Set Passwords

```
passwd fin01
# enter: fin01

passwd fin02
# enter: fin02

passwd fin03
# enter: fin03

passwd fin04
# enter: fin04

passwd fin05
# enter: fin05

passwd fin06
# enter: fin06

passwd fin07
# enter: fin07

passwd fin08
# enter: fin08

passwd fin09
# enter: fin09

passwd fin10
```

```
# enter: fin10

passwd fin11
# enter: fin11

passwd fin12
# enter: fin12

passwd fin13
# enter: fin13

passwd fin14
# enter: fin14

passwd fin15
# enter: fin15
```

Verify

```
ls /home
cat /etc/passwd | grep fin
cat /etc/group | grep finance
```

3.6 Department 6: Interns

Create Group

```
groupadd interns
```

Create Users

```
useradd -G interns interns01
useradd -G interns interns02
useradd -G interns interns03
useradd -G interns interns04
useradd -G interns interns05
useradd -G interns interns06
useradd -G interns interns07
useradd -G interns interns08
useradd -G interns interns09
useradd -G interns interns10
useradd -G interns interns11
useradd -G interns interns12
useradd -G interns interns13
useradd -G interns interns14
useradd -G interns interns15
useradd -G interns interns16
useradd -G interns interns17
useradd -G interns interns18
useradd -G interns interns19
useradd -G interns interns20
```

Set Passwords

```
passwd interns01
# enter: int01

passwd interns02
# enter: int02

passwd interns03
# enter: int03

passwd interns04
# enter: int04

passwd interns05
# enter: int05

passwd interns06
# enter: int06

passwd interns07
# enter: int07

passwd interns08
# enter: int08

passwd interns09
# enter: int09

passwd interns10
# enter: int10

passwd interns11
# enter: int11

passwd interns12
# enter: int12

passwd interns13
# enter: int13

passwd interns14
# enter: int14

passwd interns15
# enter: int15

passwd interns16
# enter: int16

passwd interns17
# enter: int17

passwd interns18
```

```
# enter: int18

passwd interns19
# enter: int19

passwd interns20
# enter: int20
```

Verify

```
ls /home
cat /etc/passwd | grep interns
cat /etc/group | grep interns
```

3.7 Department 7: IT

Create Group

```
groupadd it
```

Create Users

```
useradd -G it it01
useradd -G it it02
useradd -G it it03
useradd -G it it04
useradd -G it it05
useradd -G it it06
useradd -G it it07
useradd -G it it08
useradd -G it it09
useradd -G it it10
useradd -G it it11
useradd -G it it12
```

Set Passwords

```
passwd it01
# enter: it1

passwd it02
# enter: it2

passwd it03
# enter: it3

passwd it04
# enter: it4

passwd it05
# enter: it5

passwd it06
# enter: it6
```

```
passwd it07
# enter: it7

passwd it08
# enter: it8

passwd it09
# enter: it9

passwd it10
# enter: it10

passwd it11
# enter: it11

passwd it12
# enter: it12
```

Verify

```
ls /home
cat /etc/passwd | grep it
cat /etc/group | grep it
```

---

4) Password Policy

Password policy is configured so users cannot keep the same password forever. It improves security, supports company policy, and helps with audit readiness.

4.1 Command Meaning

```
chage -M 1 username
```

Maximum password age = 1 day. In this lab/project, it demonstrates strict password expiry enforcement.

4.2 Password Policy for Each Department

Admin Department

```
chage -M 1 adm01
chage -M 1 adm02
chage -M 1 adm03
chage -M 1 adm04
chage -M 1 adm05
chage -M 1 adm06
chage -M 1 adm07
chage -M 1 adm08
```

Server Department

```
chage -M 1 srv01
chage -M 1 srv02
chage -M 1 srv03
```

```
chage -M 1 srv04
chage -M 1 srv05
```

Sales Department

```
chage -M 1 sales01
chage -M 1 sales02
chage -M 1 sales03
chage -M 1 sales04
chage -M 1 sales05
chage -M 1 sales06
chage -M 1 sales07
chage -M 1 sales08
chage -M 1 sales09
chage -M 1 sales10
chage -M 1 sales11
chage -M 1 sales12
chage -M 1 sales13
chage -M 1 sales14
chage -M 1 sales15
chage -M 1 sales16
chage -M 1 sales17
chage -M 1 sales18
chage -M 1 sales19
chage -M 1 sales20
chage -M 1 sales21
chage -M 1 sales22
chage -M 1 sales23
```

HR Department

```
chage -M 1 hr1
chage -M 1 hr2
chage -M 1 hr3
chage -M 1 hr4
chage -M 1 hr5
chage -M 1 hr6
chage -M 1 hr7
chage -M 1 hr8
chage -M 1 hr9
chage -M 1 hr10
```

Finance Department

```
chage -M 1 fin01
chage -M 1 fin02
chage -M 1 fin03
chage -M 1 fin04
chage -M 1 fin05
chage -M 1 fin06
chage -M 1 fin07
chage -M 1 fin08
chage -M 1 fin09
chage -M 1 fin10
chage -M 1 fin11
chage -M 1 fin12
chage -M 1 fin13
chage -M 1 fin14
chage -M 1 fin15
```

Interns Department

```
chage -M 1 interns01
chage -M 1 interns02
chage -M 1 interns03
chage -M 1 interns04
chage -M 1 interns05
chage -M 1 interns06
chage -M 1 interns07
chage -M 1 interns08
chage -M 1 interns09
chage -M 1 interns10
chage -M 1 interns11
chage -M 1 interns12
chage -M 1 interns13
chage -M 1 interns14
chage -M 1 interns15
chage -M 1 interns16
chage -M 1 interns17
chage -M 1 interns18
chage -M 1 interns19
chage -M 1 interns20
```

IT Department

```
chage -M 1 it01
chage -M 1 it02
chage -M 1 it03
chage -M 1 it04
chage -M 1 it05
chage -M 1 it06
chage -M 1 it07
chage -M 1 it08
chage -M 1 it09
chage -M 1 it10
chage -M 1 it11
chage -M 1 it12
```

5) Directory Permission Setup

Department directories are created under /company. Group ownership and permissions are used to implement role-based access control.

```
mkdir -p /company/{admin,finance,hr,interns,it,sales,server,shared}

chown :admin /company/admin
chown :finance /company/finance
chown :hr /company/hr
chown :interns /company/interns
chown :it /company/it
chown :sales /company/sales
chown :server /company/server

chmod 770 /company/admin
chmod 770 /company/finance
chmod 770 /company/hr
```

```
chmod 770 /company/interns
chmod 770 /company/it
chmod 770 /company/sales
chmod 770 /company/server

chmod 1777 /company/shared
ls -ld /company/shared
```

5.1 Why chown is Used

The command chown :admin /company/admin makes admin the group-owner of /company/admin. The colon before admin means only group ownership is changed, not the user owner.

5.2 chmod 770 Meaning

- Owner: read, write, execute
- Group: read, write, execute
- Others: no permission

This ensures that users from other departments cannot access private department folders.

6) Backup Automation Using Crontab

```
mkdir -p /backup/home
crontab -e
*/30 * * * * /usr/bin/rsync -av /home/ /backup/home/ >> /var/log/home-backup.log 2>&1
```

6.1 Why Backup Every 30 Minutes?

- Minimizes data loss with a 30-minute Recovery Point Objective (RPO).
- /home contains user files, scripts, and configuration data that may change frequently.
- It balances safety and performance better than every-minute or once-a-day backups.

6.2 Why rsync -av?

- -a archive mode preserves permissions, ownership, timestamps, symbolic links, and directory structure.
- -v verbose mode helps with visibility, debugging, and logging.
- rsync performs incremental backup, so only changed files are copied.

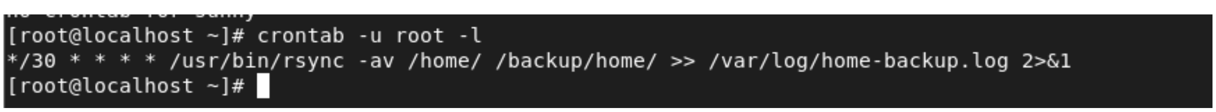


Figure 1: Crontab backup configuration proof

7) Account Expiry Management for Interns

Interns are temporary users, so their accounts are configured to expire automatically after three months.

```
usermod -e 2026-07-27 interns01
usermod -e 2026-07-27 interns02
usermod -e 2026-07-27 interns03
usermod -e 2026-07-27 interns04
```

```
usermod -e 2026-07-27 interns05
usermod -e 2026-07-27 interns06
usermod -e 2026-07-27 interns07
usermod -e 2026-07-27 interns08
usermod -e 2026-07-27 interns09
usermod -e 2026-07-27 interns10
usermod -e 2026-07-27 interns11
usermod -e 2026-07-27 interns12
usermod -e 2026-07-27 interns13
usermod -e 2026-07-27 interns14
usermod -e 2026-07-27 interns15
usermod -e 2026-07-27 interns16
usermod -e 2026-07-27 interns17
usermod -e 2026-07-27 interns18
usermod -e 2026-07-27 interns19
usermod -e 2026-07-27 interns20
```

7.1 Verification

```
chage -l interns01
chage -l interns02
```

After the expiry date, the intern account becomes inactive and cannot be used to log in.

8) Global Environment Configuration

/etc/bashrc is configured to display a global login banner and security warning to all users.

```
vim /etc/bashrc

echo "========================================"
echo " Company: TECHNOVA"
echo " User: $(whoami)"
echo " Department: $(id -gn)"
echo " Date: $(date)"
echo "----------------------------------------"
echo " Please read ~/INSTRUCTIONS.txt file"
echo " Warning: Unauthorized access is prohibited"
echo " All activities are monitored"
echo "========================================"

:wq
cat /etc/bashrc
```

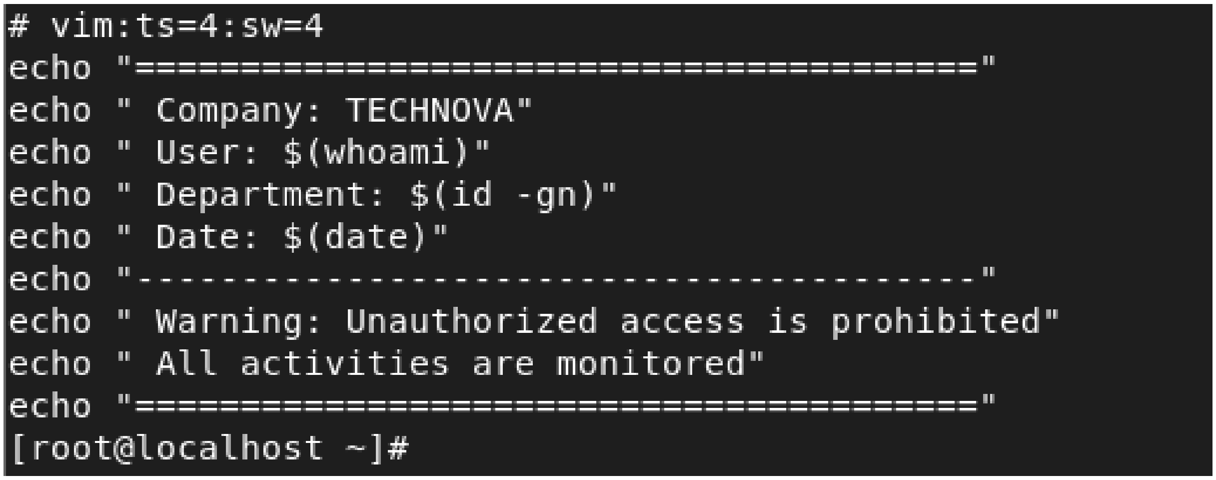


Figure 2: Global /etc/bashrc configuration proof

9) /etc/skel Configuration

/etc/skel is the default template directory. Files placed here are automatically copied into the home directory of newly created users.

```
cd /etc/skel
vim INSTRUCTIONS.txt
chmod 644 INSTRUCTIONS.txt
ls -la /etc/skel
```

9.1 INSTRUCTIONS.txt Content

```
Welcome to TECHNOVA

1) Follow company security policies
2) Do not share your password
3) Work only in your assigned directory
4) Contact admin for issues
```

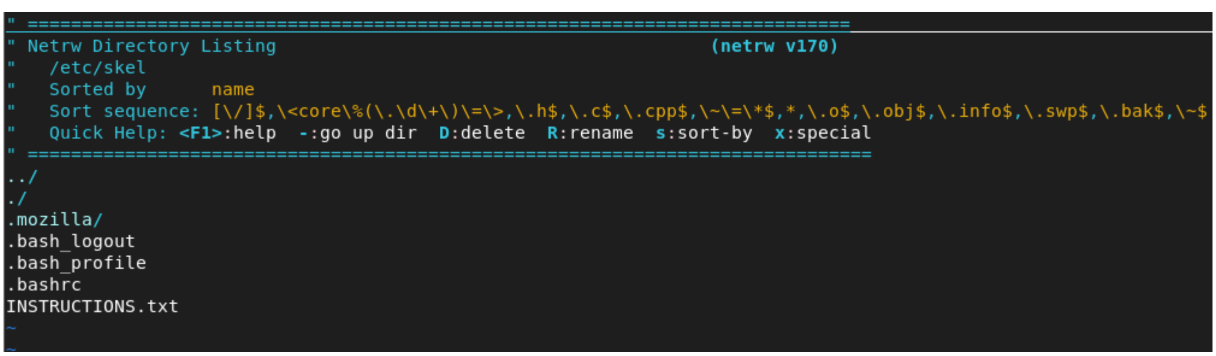


Figure 3: /etc/skel directory and INSTRUCTIONS.txt proof

10) Final Verification and Proof Screenshots

```
ls /home
cat /etc/passwd
```

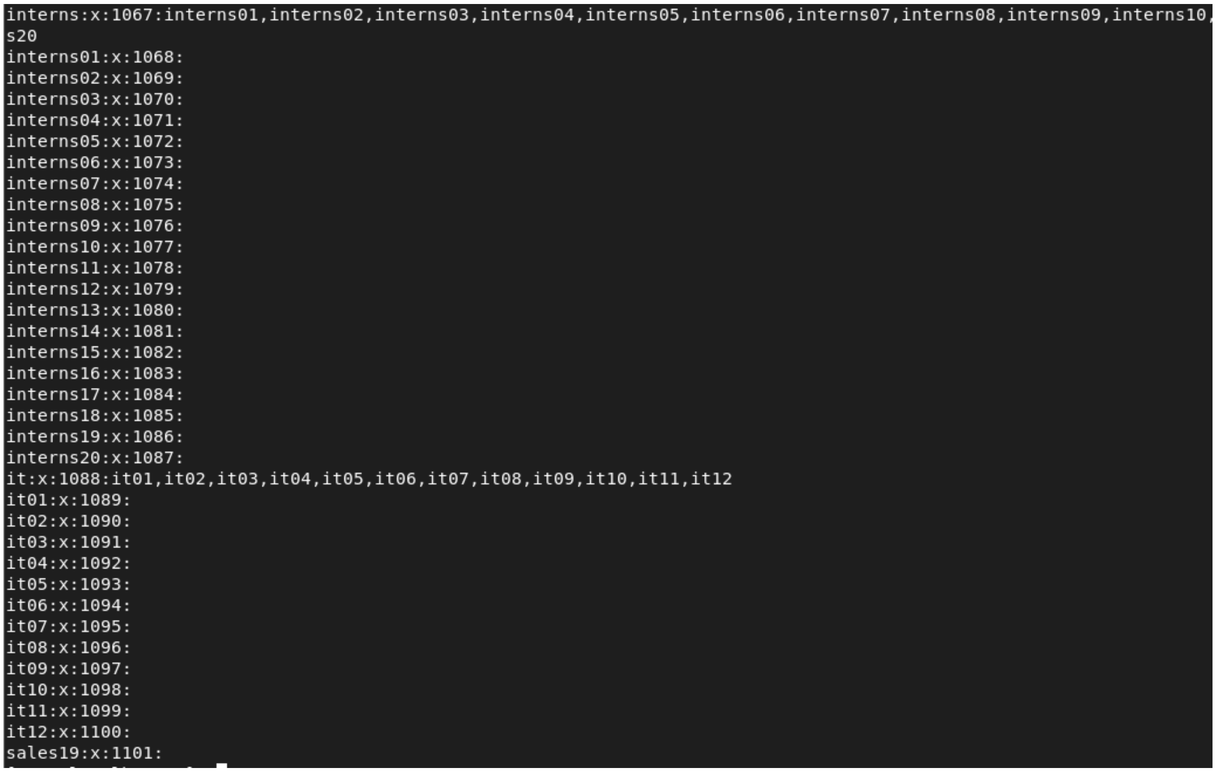
```
cat /etc/group
id adm01
id sales01
id srv01
id fin01
id hr1
id interns01
id it01
```

```
[root@localhost ~]# ls /home
adm01  fin04  fin15  interns01  interns12  it03  sales01  sales12  sales23
adm02  fin05  hr1    interns02  interns13  it04  sales02  sales13  srv01
adm03  fin06  hr10   interns03  interns14  it05  sales03  sales14  srv02
adm04  fin07  hr2    interns04  interns15  it06  sales04  sales15  srv03
adm05  fin08  hr3    interns05  interns16  it07  sales05  sales16  srv04
adm06  fin09  hr4    interns06  interns17  it08  sales06  sales17  srv05
adm07  fin10  hr5    interns07  interns18  it09  sales07  sales18  sunny
adm08  fin11  hr6    interns08  interns19  it10  sales08  sales19
fin01  fin12  hr7    interns09  interns20  it11  sales09  sales20
fin02  fin13  hr8    interns10  it01       it12  sales10  sales21
fin03  fin14  hr9    interns11  it02       sai   sales11  sales22
[root@localhost ~]#
```

Figure 4: /home directory users proof

```
hr2:x:1003:1004::/home/hr2:/bin/bash
hr3:x:1004:1005::/home/hr3:/bin/bash
hr4:x:1005:1006::/home/hr4:/bin/bash
hr5:x:1006:1007::/home/hr5:/bin/bash
hr6:x:1007:1008::/home/hr6:/bin/bash
hr7:x:1008:1009::/home/hr7:/bin/bash
hr8:x:1009:1010::/home/hr8:/bin/bash
hr9:x:1010:1011::/home/hr9:/bin/bash
hr10:x:1011:1012::/home/hr10:/bin/bash
fin01:x:1012:1014::/home/fin01:/bin/bash
fin02:x:1013:1015::/home/fin02:/bin/bash
fin03:x:1014:1016::/home/fin03:/bin/bash
fin04:x:1015:1017::/home/fin04:/bin/bash
fin05:x:1016:1018::/home/fin05:/bin/bash
fin06:x:1017:1019::/home/fin06:/bin/bash
fin07:x:1018:1020::/home/fin07:/bin/bash
fin08:x:1019:1021::/home/fin08:/bin/bash
fin09:x:1020:1022::/home/fin09:/bin/bash
fin10:x:1021:1023::/home/fin10:/bin/bash
fin11:x:1022:1024::/home/fin11:/bin/bash
fin12:x:1023:1025::/home/fin12:/bin/bash
fin13:x:1024:1026::/home/fin13:/bin/bash
fin14:x:1025:1027::/home/fin14:/bin/bash
fin15:x:1026:1028::/home/fin15:/bin/bash
srv01:x:1027:1030::/home/srv01:/bin/bash
srv02:x:1028:1031::/home/srv02:/bin/bash
srv03:x:1029:1032::/home/srv03:/bin/bash
srv04:x:1030:1033::/home/srv04:/bin/bash
srv05:x:1031:1034::/home/srv05:/bin/bash
adm01:x:1032:1036::/home/adm01:/bin/bash
adm02:x:1033:1037::/home/adm02:/bin/bash
adm03:x:1034:1038::/home/adm03:/bin/bash
adm04:x:1035:1039::/home/adm04:/bin/bash
adm05:x:1036:1040::/home/adm05:/bin/bash
adm06:x:1037:1041::/home/adm06:/bin/bash
adm07:x:1038:1042::/home/adm07:/bin/bash
adm08:x:1039:1043::/home/adm08:/bin/bash
sales01:x:1040:1045::/home/sales01:/bin/bash
sales02:x:1041:1046::/home/sales02:/bin/bash
sales03:x:1042:1047::/home/sales03:/bin/bash
sales04:x:1043:1048::/home/sales04:/bin/bash
```

Figure 5: /etc/passwd user verification proof



```
interns:x:1067:interns01,interns02,interns03,interns04,interns05,interns06,interns07,interns08,interns09,interns10,
s20
interns01:x:1068:
interns02:x:1069:
interns03:x:1070:
interns04:x:1071:
interns05:x:1072:
interns06:x:1073:
interns07:x:1074:
interns08:x:1075:
interns09:x:1076:
interns10:x:1077:
interns11:x:1078:
interns12:x:1079:
interns13:x:1080:
interns14:x:1081:
interns15:x:1082:
interns16:x:1083:
interns17:x:1084:
interns18:x:1085:
interns19:x:1086:
interns20:x:1087:
it:x:1088:it01,it02,it03,it04,it05,it06,it07,it08,it09,it10,it11,it12
it01:x:1089:
it02:x:1090:
it03:x:1091:
it04:x:1092:
it05:x:1093:
it06:x:1094:
it07:x:1095:
it08:x:1096:
it09:x:1097:
it10:x:1098:
it11:x:1099:
it12:x:1100:
sales19:x:1101:
```

Figure 6: /etc/group department mapping proof

11) Conclusion

This project demonstrates a complete Linux administration workflow for TECHNOVA. It covers department-wise user and group creation, password configuration, password policy, role-based access control, automated backup, intern account expiry, global login banners, /etc/skel onboarding, and verification using real terminal screenshots.

11.1 Project Outcome

- Centralized user and group management completed.
- Department-wise access control implemented.
- Password and account lifecycle policies applied.
- Automated backup configured using cron and rsync.
- Default onboarding files configured using /etc/skel.
- Screenshots added as implementation proof.